

Integrating Social Context Into EmoBertGCN for Sentiment Analysis of Filipino Political Discourse

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Abstract—With the rise of political tensions both internationally and locally in the Philippines, the analysis of online political discourse becomes a source of crucial information to derive public opinion towards government policies, with elections serving as the primary context for sentiment extraction. However, political events often carry polarizing opinion which create imbalanced datasets, which remains one of the major hurdles in sentiment analysis. This study aims to address that concern by extending the BertGCN model by injecting conversational context into the graph structure of its GCN component, enabling sentiment to propagate across reply chains grounded in Emotional Contagion theory and the principle of Homophily. Using a corpus of 73,799 tweets from the 2025 Philippine general midterm elections, the proposed model was evaluated against a RoBERTa baseline fine-tuned for code-switched Tagalog sentiment and the original EmoBertGCN. Ten-fold cross-validation shows macro F1 scores of 85.3% (RoBERTa baseline), 84.6% (BertGCN baseline), and 84.9% (extended BertGCN), indicating that the proposed integration yields no significant performance gains ($p > 0.05$ for both comparisons). Despite the null finding, this research contributes a manually and machine-annotated dataset of 73,799 election-period tweets as a resource for future Tagalog NLP research. A BERT-LDA topic modelling approach was then utilized to yield distinct topic clusters for deeper sentiment analysis. The study contributes new datasets in Filipino as well as provides methodological insights for future graph-based sentiment analysis in low-resource, conversation-sparse environments.

Index Terms—Sentiment analysis, Graph neural networks, EmoBertGCN, Filipino political discourse, Low-resource languages, Class imbalance, Social context

I. INTRODUCTION

Natural Language Processing (NLP) has evolved from the rule-based and keyword-driven systems into sophisticated models capable of understanding the tone, context, and emotion in text. Among the many categories of NLP, sentiment analysis has been pivotal in understanding and gauging human sentiment towards specific topics/subjects. This specific branch of NLP provides researchers, organizations, companies, and governments with meaningful information that could improve their services, projects, or processes. In recent years, the sentiment classification task has been powered by deep learning and transformer models, which include all the Bidirectional Encoder Representations from Transformers (BERT) variants [1]. These transformer models have achieved significantly improved accuracies and are now deemed state-of-the-art models. However, these new models are still limited by the amount of corpus there is for the specific language under training.

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Languages that lack large and well-annotated datasets are called low-resource languages, which these new models have difficulty understanding.

Despite the effectiveness of transformer-based models, recent literature has pushed for new frameworks to improve sentiment classification. Some of these approaches now include the entire corpus when classifying sentiment. They now employ the use of Graph-based learning, specifically Graph Neural Networks (GNNs), to take advantage of the entire corpus when classifying individual documents. Foundational models such as BertGCN [2], preceded by TextGCN [3], combine graph-structured learning with transformer embeddings to model semantic relationships across large corpora. However, despite these advancements, these new frameworks still lack the conversational context that social media posts contain.

This research seeks to extend the EmoBertGCN framework by including social interaction edges, specifically conversation-based relationships. In doing so, the sentiment analysis framework gains even more context. Using the 2025 Philippine general midterm elections as the corpus to train on, the research also provides valuable resources and insights for future NLP research in the context of the Philippines. As a result, the study then provides not only a methodological improvement upon previous iterations of the BertGCN, but also a deeper understanding of the political expression of Filipinos regarding local issues in online spaces.

A. Statement of the Problem

Despite advancements in transformer-based sentiment analysis through the procurement of more quality Tagalog corpora, pre-training of more language models, and integration of Graph Neural Networks in sentiment classification, several critical challenges still remain:

- 1) Sentiment imbalance in political datasets restricts skew model training and leads to biased and unreliable sentiment prediction.
- 2) Current GNN-integrated sentiment models (BertGCN) capture document-word and word-word relationships, but ignore social relationships between posts. Consequently, the framework fails to take advantage of emotional propagation and the interactional context within social media conversations.
- 3) There is limited research applying these newly pre-trained language models and advanced architectures to Philippine-specific datasets, especially ones situated during electoral periods, which would provide significant political and social insight towards public sentiment.

B. Research Objectives

The objectives of this research can be broken down into the following goals. The first objective is determine if the inclusion of conversational context in sentiment analysis of political discourse would improve model performance in low-resource and class-imbalanced datasets. The second objective is to create more resources for the application of NLP in Tagalog Corpora through the creation of a labeled resource on the Philippine 2025 general midterm elections for further research to better capture and understand the context during that key political event.

C. Significance of the Study

This research is significant in both NLP tasks and socio-political analysis. In the natural language processing domain, this research advances sentiment analysis for low-resource languages by extending the EmoBertGCN framework to incorporate social interaction edges. In doing so, this framework addresses both the lack of data and pretraining for Tagalog- and Filipino-code-mixed text and the incomplete representation of relationships among social media posts. In the socio-political domain, this paper provides a foundation for examining online political discourse in the Philippines. Consequently, this research is beneficial to computational linguists, social scientists, and policymakers concerned with dynamic public sentiment in online spaces. In relation to the United Nations Sustainable Development Goals (SDGs), this study addresses two of the goals. First is SDG 9 (Industry, Innovation, and Infrastructure), as the study aims to strengthen existing sentiment analysis methods to capture fully and accurately the public political sentiment of Filipinos. This also relates to SDG 16 (Peace, Justice, and Strong Institutions), as the results of this study would better equip political leaders to improve and adapt their governance to actual public sentiment.

D. Scope and Limitations

This study focuses on sentiment analysis of Twitter (X) data, which is situated during the 2025 Philippine general midterm elections. The scope includes text in Tagalog, English, or code-mixed formats. Due to this, results may not be generalized to other social media platforms or key political events. The methodology only covers the integration of social interaction edges into EmoBertGCN. It also includes evaluation using quantitative performance metrics and topic modelling for interpretability of model inference. The implementation of social edges is limited to only conversation-based relationships. This means posts under a conversation thread. It does not consider the temporal attribute of the posts, interaction metrics, or the unique users tied to individual posts in the dataset. This decision was made to isolate the impact of conversational structure itself, independent of user identity or timing, which can be explored in future work. The anonymization of unique user identifiers is done to maintain data privacy.

II. RELATED STUDIES

A. Sentiment Analysis

In the context of sentiment analysis of Filipino content, mBERT is able to perform reasonably well, given its 104-language training enables zero-shot transfer between English and Tagalog within the same sentence. In a study conducted by Maceda et al. [4], they demonstrated mBERT's promising performance with an accuracy and F1 score of 80.21% and 81.14% with only minimal pre-processing when handling code-mixed text (Tagalog and English). While mBERT provides a solid baseline, it still struggles with languages that are identified as low-density, less commonly taught, under-resourced, and low-resource in nature, which Tagalog falls under [5], [6].

Despite the gains made possible by the introduction of new Tagalog pre-trained models and corpus texts, a fundamental issue remains which is class imbalance. Class imbalance happens when the distribution of class labels favors one or more majority classes [7]. This issue becomes even more obvious in political datasets. According to Schöne et al. [8], tweets with negative emotional language were considerably more likely to be retweeted or shared than those with positive language. Positive language, on the other hand, was a much weaker predictor than negative language even in the positive contexts. Consequently, the minority class, which in this case is the positive sentiment class, is systematically underrepresented, leading to lower per-class performance on those labels even when overall accuracy appears acceptable [9].

Some common solutions to class imbalance include resampling, data augmentation, and modifying the loss function [7]. While effective in certain settings, these strategies address the frequency of minority-class samples without utilizing the social media structure surrounding those samples. A minority-class document, regardless of how many times it is resampled, still carries only its own local context when processed by a sequential model such as BERT. The model has no mechanism to draw on the broader distributional patterns of the corpus when classifying any individual document.

Graph-based frameworks such as TextGCN and BertGCN offer a structurally different solution to this problem. By representing the entire corpus as a heterogeneous graph and performing node classification through message passing, these models allow individual documents to aggregate information from connected nodes through edges based on each document's lexical features. This means that even a minority-class document benefits from the collective signal present in its graph neighborhood, rather than being classified purely in isolation. Yao et al. [3] and Liu et al. [2] both note that their respective frameworks achieve strong performance even under limited training data conditions, which is attributable in part to this corpus-wide information sharing. In this sense, graph-based learning offers a representational solution to class imbalance that complements, rather than replaces, data-level mitigation strategies.

B. Graph-Based Text Classification Methods

In an effort to improve sentiment analysis, frameworks that include Graph Neural Networks (GNNs) have been introduced. A comprehensive review conducted by Wang et al. [10] states that while numerous recent text classification models applied the sequential deep learning technique (methods that process text information in order, such as RNN, LSTM), graph neural network-based models can directly deal with complex structured text data and exploit global information that may be overlooked by sequential models.

A notable example of GNN being used for text classification is in the study entitled “Graph Convolutional Networks for Text Classification” [3], which was proposed with the intention of designing a model trained on the entirety of the corpus represented as a heterogeneous graph where words and documents are represented as nodes. TextGCN achieves state-of-the-art results on several benchmarks without relying on pre-trained word embeddings, demonstrating robustness when faced with limited training data.

TextGCN is later built upon by BertGCN, proposed by Lin et al. [2]. They proposed a hybrid model that combines the large-scale pretraining of BERT models with the transductive learning of GCN. Extending upon TextGCN, BertGCN constructs the heterogeneous graph where the document nodes are initialized using BERT embeddings rather than an identity matrix. Other characteristics of the graph structure, such as the word-document edges and the word-word edges (which now use a Positive Pointwise Mutual Information (PPMI)), are mirrored in this new proposed hybrid framework. BertGCN achieves state-of-the-art results on several text classification benchmarks, especially on longer documents where the GCN component provides significant gains.

A more recent and relevant study in the Philippine context by Aquino et al. [11] directly addresses the methodological gaps in sentiment analysis for low-resource languages, specifically Tagalog. The authors proposed the EmoBertGCN hybrid model, which employs the use of the previously proposed BertGCN but now incorporates emojis, hashtags, and user mentions to improve semantic understanding on online sentiment. In this approach, they treated these tokens as essential features for semantic and sentiment understanding rather than noise to be filtered out during the preprocessing stage. They employed the use of their model on an imbalanced dataset of Filipino political tweets in which they achieved a macro F1-score of 68.72%, outperforming a wide range of traditional machine learning, deep learning, and transformer-based baseline models. Furthermore, they extended their text classification task by introducing a topic modeling framework that combines BERT embeddings with Latent Dirichlet Allocation (LDA) and Log-Likelihood Ratio (LLR) to yield more distinct topic clusters for deeper sentiment analysis. However, now that BertGCN and EmoBertGCN demonstrate state-of-the-art performances, a remaining challenge is the implementation of document-to-document relationships, especially in the context of applying these hybrid sentiment analysis frameworks on social media.

C. Emotional Contagion

This study builds upon the discussed Bert and EmoBertGCN hybrid models using the idea of emotion propagating in social networks. In doing so, the hybrid model is provided with more context to work with when learning the relationships between not only documents and words and words to words, but also between documents. Emotional contagion is defined by the transfer of emotional states from person to person, leading individuals to experience the same emotion as the people around them [12], [13]

D. Assumption of Homophily

Building upon the proof of emotional contagion in online social platforms, studies have also shown that similar thinking individuals tend to group themselves together [14]. This pattern remains true even in online political discourse (specifically in Twitter, now known as X), as the study by Halberstam & Knight [15] establishes that homophily and group size of similar-thinking users systematically bias the flow of political information on social media. Furthermore, this concept of homophily has also been used to describe GNNs as a prevalent characteristic, often implied, in many foundational GNN architectures, which is their assumption of homophilous graph structures. This can be seen with the use of aggregation schemes, which work well when connected nodes share similar features and labels [16].

Despite the dominant homophily characteristic of social networks online, it is also important to note that, though homophily dominates social networks, heterophily may still exist. Studies like that of De Francisci Morales et al. [17] show that cross-cutting interactions happen even when politically segregated communities exist. Though De Francisci Morales et al.’s study is situated in a different media platform (Reddit), this demonstrates the existence of these types of interactions. Given this understanding of the social network structure, it is also important to discuss the clear struggle that GNNs face when handling heterophilous graphs [16], indicating that heterophily remains a fundamental challenge in state-of-the-art GNN models.

E. Research Gap

BertGCN and EmoBertGCN integrate document–word and word–word relationships but ignore relationships between documents. This limits their ability to capture interactional context on social media, where posts are connected via replies and conversation threads. The inclusion of these conversation-thread relationships is grounded in the previously discussed theories of emotional contagion, which states that the emotional states can transfer between users via exposure [13], [18].

By integrating conversation-based edges, the framework aims to construct a graph that faithfully represents the interactional dynamics present within the social media environment. The integration of these conversation edges when constructing the corpus-level graph will enable the BertGCN to realistically capture the sentiment using conversational context by providing a pathway for the GCN to learn relationships between

posts. However, it is important to note that the addition of the conversation-based edges may encode both homophilous and heterophilous interactions as established in other social media platforms like Reddit [17]. Therefore, the integration of these conversation edges when constructing the corpus-level graph will enable the BertGCN to realistically capture political discourse sentiment.

III. METHODS

A. Data Preparation

Tweets were collected from December 24, 2024 up until May 31, 2025 in the context of the 2025 Philippine midterm elections with the use of a 3rd party service named twit-terapi.io. The following keywords were used in scraping the tweets:

- 1) *Bobotante, Botante, Botohan, Bumoto, Comelec, Commission on Elections, Eleksyon, Eleksyon2025, Halalan, Halalan2025, Iboto, Kandidato, Philippine Elections, Philippine Politics, Senado, Senador.*

The collection was done every 12:00am Philippine Standard Time (PST, UTC+8) and only captures tweets created within the past 24 hours. This process is repeated daily. The following data were collected: Tweet Content, Engagement Metrics (such as Retweet count, Reply count, Like count, Quote count, View count, Bookmark count), Date posted, Author Username, Username of the person being replied to, Conversation ID. All tagged usernames (ex: @username) and forms of IDs are replaced with a unique hashid value based on original value retaining data integrity.

The dataset was filtered to only include posts that contained Tagalog content. This was done using the text classification tool fastText [19] without additional fine-tuning to accurately identify and incorporate only tweets that include Tagalog text. After filtering the corpus, the dataset size was reduced from 217,640 to 123,864 tweets. Preprocessing followed Cruz & Cheng [20] and Aquino et al. [11], including URL replacement, mention/@ retention, punctuation normalization, ASCII filtering, and emoji text conversion. In the creation of the labeled set, a group of native speakers annotated a stratified 5% sample using guidelines Table I validated by an expert. Inter-annotator agreement (Cohen's $\kappa = 0.77$) indicates the experiment results use for tentative conclusion.

Prior to the experiment, an examination of the dataset's graph structure was conducted. It was revealed that the dataset features sparse conversation connectivity in the training set. Only 1,051 of 4,464 posts (23.5%) participated in conversation threads with two or more posts. The remaining 3,413 posts (76.5%) existed as isolated nodes with no conversation edges. The 376 conversation threads consisted of about 2.8 posts each, indicating mostly shallow interaction chains. However, among the few posts that are connected, most of them are homophilic in nature. Approximately 97.2% of the subset of connected tweets are homophilic in nature. The negative sentiment class constitutes a majority of the connected posts, with around 10% of negative posts being connected to other posts with differing sentiment labels. Although the positive class has the highest percentage of heterophilic connected

TABLE I: Annotation Guidelines for Sentiment Classification

Label	Sentiment	Description
0	POSITIVE	The speaker is using positive language toward a candidate, platform, or election-related topic. Examples: support, admiration, enthusiasm, forgiveness, fostering success.
1	NEGATIVE	The speaker is using negative language toward a candidate, platform, or election-related topic. Examples: criticism, pessimism, questioning validity/competence, failure, sarcasm, ridicule or mockery.
2	NEUTRAL	The speaker does not use discernible sentiment or conveys information/facts about the 2025 elections. May contain balanced positive/negative sentiments about the primary subject.

posts, its actual number of connected posts is significantly lower than that of the negative and neutral classes, with only 503 connected posts. These findings indicate that the risk of heterophilic posts would be insignificant towards the training of the GCN component of the hybrid model. However, the sparsity of the connected posts posed a new risk of insignificant returns from integrating conversation edges between posts. Since the dataset came from a public repository, conversation connectivity was a natural feature of the data, not something that could be adjusted. Therefore, the experiment proceeded with sparse data, reflecting a realistic scenario where conversation data may be limited.

B. Data Training and Inference

Following Aquino et al. [11], this study extends the **EmoBertGCN** architecture by incorporating **social interaction edges** into the graph structure, as shown in Figure 1. The original model integrates document-word (TF-IDF) and word-word (PPMI) relations. This study extended the EmoBertGCN model by adding a third edge type: document-to-document edges, defined by reply relationship. Heterogeneous graph $G = (V, E)$ was constructed, where Equation. 1 describes the nodes and Equation. 2 describes the edges to be implemented. The feature of a document node is defined in Equation. 3 where the document embedding was obtained from a BERT model, while the feature of the word node is initialized to zero. The weight of the edges are further defined in Equation. 4 which now includes the social edges. The output feature matrix is then calculated for the i -th layer of the GCN, $L^{(i)}$, using Equation. 5 where ρ is an activation function (ReLU), $\hat{A}$ is the normalized adjacency matrix, and $W^{(i)}$ is a trainable weight matrix for the i -th layer with dimensions $R^{d_{i-1} \times d_i}$.

$L^{(0)} = X$ is the input feature matrix of the GCN while the outputs are treated as the final representation for documents which is fed to a softmax layer for classification. Additionally, the document embeddings are then fed directly to a dense layer with softmax activation. Afterwards, the predictions of both the BERT and GCN components are linearly interpo-

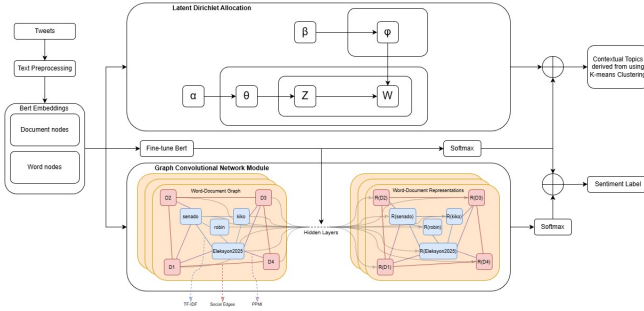


Fig. 1: The proposed architecture (extension of EmoBertGCN)

lated to obtain the final classification which is described in Equation. 6. For the BERT model, ‘dost-asti/Roberta-tl-sentiment-analysis’ [21] is selected over ‘jcbllaise/roberta-tagalog-base’ [20] as it is specifically pre-trained for sentiment analysis on Filipino political sentiment in social media.

For the hyperparameters, a feature dimension of 768 was employed to match the RoBERTa-base encoder output, with a GCN hidden layer of 200 dimensions and a dropout rate of 0.5 for regularization. For optimization, the Adam optimizer is employed with learning rates of 1×10^{-5} for BERT and 1×10^{-3} for GCN components. Training utilized a batch size of 32 over 5 epochs, with early stopping patience of 3 epochs. The interpolation parameter λ was set to 0.7 based on Lin et al. [2]’s findings. The maximum sequence length for tokenization was set to 128 tokens. Furthermore, a pre-finetuning strategy with 10 epochs at 2×10^{-5} learning rate before joint training.

$$V = V_{\text{doc}} \cup V_{\text{word}} \quad (1)$$

$$E = E_{\text{TF-IDF}} \cup E_{\text{PPMI}} \cup E_{\text{social}} \quad (2)$$

$$X = \begin{pmatrix} X_{\text{doc}} \\ 0 \end{pmatrix}_{(n_{\text{doc}} + n_{\text{word}}) \times d} \quad (3)$$

$$A_{ij} = \begin{cases} \text{PPMI}(i, j), & \text{if } i, j \in V_{\text{word}}, i \neq j \\ \text{TF-IDF}(i, j), & \text{if } i \in V_{\text{doc}}, j \in V_{\text{word}} \\ 1, & \text{if } i, j \in V_{\text{doc}} \text{ and social edge exists} \\ 1, & \text{if } i = j \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

$$L^{(i)} = \rho(\tilde{A}L^{(i-1)}W^{(i)}) \quad (5)$$

$$Z = \lambda Z_{\text{GCN}} + (1 - \lambda)Z_{\text{BERT}} \quad (6)$$

C. Experimental Validation

Experimental validation followed 10-fold cross-validation with macro F1-score as the primary evaluation metric. Within each fold, the training pool was further split into training (85%) and validation (15%) sets. Conversation threads were preserved across folds to prevent data leakage. Macro-averaged F1-score was used as the primary metric, consistent

with established practices in sentiment analysis evaluation [9], [22], [23]. In determining the significance of the results, a one-tailed t-test was conducted to assess whether our model outperforms the baseline models, which are listed as follows:

- RoBERTa Sentiment [21]
- EmoBertGCN [11]
- Our Proposed Model (EmoBertGCN with Social Edges)

D. Topic Modeling

The study employed the use of Aquino et al.’s **Emotion-based Topic Modeling using a Hybrid model of BERT and LDA (Latent Dirichlet Allocation)** to assess the sentiments of all the tweets regarding the Philippine general midterm elections. BERT provides the contextual, syntactic, and semantic meaning of the text, while LDA acts as a thematic and topical feature extractor by providing the topic distribution of each document. When combined, it is employed to reveal unseen topics within each sentiment category in the dataset. Given that concatenating BERT embeddings and LDA topic distribution vector results in a high-dimensional vector, it must be reduced in order to better visualize and cluster the topics. Hence, Uniform Manifold Approximation and Projection (UMAP) was used to reduce its dimensions [24]. Afterwards, K-means clustering was used to derive the topic clusters based on the output of the BERT-LDA hybrid model. UMass Coherence Index (UCI) and UMass Coherence Scores was computed to objectively measure the interpretability of the topics derived from the K-means clustering step, as shown in Equation. 7 and Equation. 8, respectively [25]. Lastly, A log-likelihood ratio Equation. 9, which was adopted from the methodology of Chang et al. [26], was used to extract sentiment words to better understand the dataset’s intrinsic pattern by creating word clouds.

$$C_{UCI} = \frac{2}{N(N-1)} \sum_{i=1}^{N-1} \sum_{j=i+1}^N \log \frac{P(w_i, w_j) + \epsilon}{P(w_i) \cdot P(w_j)} \quad (7)$$

$$C_{UMass} = \frac{2}{N(N-1)} \sum_{i=1}^{N-1} \sum_{j=i+1}^N \log \frac{P(w_i, w_j) + \epsilon}{P(w_j)} \quad (8)$$

$$LLR(w, T) = 2 \cdot \log \left(\frac{p(w|T)^k (1 - p(w|T))^m}{p(w)^{k+l} (1 - p(w))^{m+n}} \right) \times p(w|-T)^l (1 - p(w|-T))^n \quad (9)$$

IV. RESULTS AND DISCUSSION

A. Model Evaluation

This chapter discusses the results in two main sections. The first section discusses the training and evaluation of the three models: a RoBERTa baseline, the BertGCN baseline, and the extended BertGCN with conversation edges. The models were evaluated using a 10-fold cross validation on a labeled dataset. Despite the intuition that conversation structure could mitigate class imbalance, statistical analysis reveals no significant performance gain from the graph-based models. The

second section applies the best-performing model to infer sentiment on 69,335 unlabeled tweets, revealing a dominant negative sentiment class. Topic modeling using BERT-LDA further identifies five interpretable thematic clusters during the Philippine election period.

Table II shows the full metrics of all the models evaluated in the experiment. All three models achieved comparable macro F1 scores: RoBERTa baseline (0.8534 ± 0.0293), BertGCN without conversation edges (0.8456 ± 0.0253), and BertGCN with conversation edges (0.8487 ± 0.0254). The relatively high standard deviation values across all metrics indicate that the performance of each model oscillates significantly per fold, suggesting that the performance of all models was sensitive to the composition of the folds.

TABLE II: Per-class and macro-averaged performance of the three models over 10-fold cross-validation (mean $\pm$ std).

Class	Metric	RoBERTa	BertGCN (no conv)	BertGCN (+ conv)
Positive	Precision	0.7042 $\pm$ 0.0772	0.7833 $\pm$ 0.0831	0.7545 $\pm$ 0.0869
	Recall	0.7317 $\pm$ 0.0632	0.6394 $\pm$ 0.0742	0.6752 $\pm$ 0.0840
	F1	0.7155 $\pm$ 0.0586	0.6984 $\pm$ 0.0463	0.7050 $\pm$ 0.0442
Negative	Precision	0.9237 $\pm$ 0.0305	0.9045 $\pm$ 0.0371	0.9121 $\pm$ 0.0286
	Recall	0.9357 $\pm$ 0.0194	0.9484 $\pm$ 0.0245	0.9362 $\pm$ 0.0337
	F1	0.9294 $\pm$ 0.0191	0.9254 $\pm$ 0.0213	0.9236 $\pm$ 0.0231
Neutral	Precision	0.9334 $\pm$ 0.0284	0.9228 $\pm$ 0.0375	0.9234 $\pm$ 0.0351
	Recall	0.8996 $\pm$ 0.0469	0.9065 $\pm$ 0.0511	0.9132 $\pm$ 0.0428
	F1	0.9152 $\pm$ 0.0238	0.9131 $\pm$ 0.0249	0.9175 $\pm$ 0.0279
Macro	Precision	0.8538 $\pm$ 0.0308	0.8702 $\pm$ 0.0304	0.8633 $\pm$ 0.0315
	Recall	0.8556 $\pm$ 0.0317	0.8314 $\pm$ 0.0291	0.8415 $\pm$ 0.0279
	F1	0.8534 $\pm$ 0.0309	0.8456 $\pm$ 0.0267	0.8487 $\pm$ 0.0268
Accuracy	F1	0.9000 $\pm$ 0.0251	0.8980 $\pm$ 0.0242	0.8982 $\pm$ 0.0268
Weighted	Precision	0.9031 $\pm$ 0.0222	0.8986 $\pm$ 0.0219	0.8996 $\pm$ 0.0221
	Recall	0.9000 $\pm$ 0.0251	0.8980 $\pm$ 0.0242	0.8982 $\pm$ 0.0268
	F1	0.9006 $\pm$ 0.0243	0.8957 $\pm$ 0.0240	0.8971 $\pm$ 0.0253

The performance metrics also indicate that all three models demonstrated strong and consistent performance on the negative and neutral classes, with macro F1 scores of around 0.92 and 0.91, respectively, across all configurations can be seen in Table II. The consistency of results illustrates that all 3 models were able to effectively learn these two classes. This is further reflected in the overall confusion matrices as seen in Table III where the models were able to correctly classify a majority

TABLE III: Aggregated confusion matrices over 10-fold cross-validation for all three models.

	RoBERTa (Baseline)			BertGCN (no conv)			BertGCN (+ conv)		
	Pos	Neg	Neu	Pos	Neg	Neu	Pos	Neg	Neu
Pos	369	88	46	321	118	64	340	107	56
Neg	84	2140	62	52	2169	65	76	2139	71
Neu	76	92	1507	43	114	1518	47	99	1529

Note: Rows represent true labels; columns represent predicted labels.

of the negative and neutral samples compared to the positive samples.

In contrast, all models performed considerably worse on the positive class, with the macro F1 scores of 0.70 across all models and all folds. As shown in Table III, the positive class shows the most misclassifications as either the negative or neutral classes, while the reverse (negative and neutral samples being misclassified as positive) occurred significantly less. This asymmetric pattern is consistent with the class imbalance present in the dataset since the model will better learn to classify the class/es with the most documents. Consequently, these results suggest that, since it is consistent across all three models, class imbalance is a more dominant influence on performance than any architectural differences between models.

This is further validated by the results of the one-tailed t-test gathered from the experiments. Statistical significance was assessed using exact joint permutation tests following the Westfall–Young procedure. Since both comparisons share the BertGCN (+conv, $m = 0.7$) scores as a common group, the standard Bonferroni correction would be over-conservative due to the positive correlation between tests. With $n = 10$ folds, all 184,756 combinations were enumerated exhaustively, yielding exact (non-approximate) marginal and joint p -values. The balance parameter $m = 0.7$ was pre-specified following Lin et al. [2]. All tests are one-tailed at $\alpha = 0.05$.

As shown in Table IV, neither comparison showed a significant performance increase. The proposed BertGCN with conversation edges did not significantly outperform the RoBERTa baseline ($p = 0.642$, $d = -0.163$) or the BertGCN baseline ($p = 0.401$, $d = 0.114$). The joint family-wise error rate ($p = 0.647$) further confirms that the family of tests produced no significant result. Lastly, the small effect sizes confirm that the models' performances are comparable.

The insignificant results may be attributed to the sparsity of conversation-connected posts in the dataset. The addition of conversation edges is premised on the assumption that directly connected posts carry a meaningful relational signal that can inform sentiment classification to be robust against class imbalance issues. However, when the proportion of such connections is low, the graph component has limited opportunity to propagate class-discriminative information across posts, and the added edges may contribute insufficient signal to produce a measurable difference in performance. As a result, the GNN component had little to learn from the edges connecting posts within the same conversation. This provides a structural explanation for the lack of significant performance

TABLE IV: Statistical comparison of macro F1 scores across 10-fold cross-validation.

Comparison	Mean A	Mean B	Cohen's d	p -value	Sig.
BertGCN (+conv) vs. RoBERTa	0.8487	0.8534	-0.163	0.642	No
BertGCN (+conv) vs. BertGCN (no conv)	0.8487	0.8456	+0.114	0.401	No
Joint FWER (Westfall-Young)				0.647	No

Note: Exact permutation tests with Westfall-Young joint FWER correction. All tests are one-tailed (hypothesis: BertGCN with conv > baseline).

gains over the baseline BertGCN.

The lack of performance gain of the BertGCN over the RoBERTa baseline contrasts with the results of Aquino et al. [11]'s experiments, which were under more severe class imbalance conditions. This case is most noticeable with their neutral class, which only comprised 7.3% of samples (583 out of 8,035), which, when compared to the class imbalance issue of this study, is more severe than the dataset used in this study. Furthermore, the BERT component model utilized in Aquino et al. [11]'s paper was a general-purpose Filipino pretrained language model. On the other hand, this study employed the use of a Filipino pretrained RoBERTa model that is specialized for sentiment classification on code-switched Tagalog text [21]. This difference in pretraining provided the models used in this study with stronger sentiment-relevant representations, raising the performance ceiling and further limiting any room for improvement provided by the graph component of BertGCN.

The posited improved towards the positive class is insignificant across any configuration. All three models yielded nearly identical positive-class F1 scores (≈ 0.70), indicating that message-passing did not meaningfully enrich representations of positive-class instances. Two confounding factors prevented a fair test of the underlying hypothesis. First, the specialized RoBERTa backbone (fine-tuned on code-switched Tagalog sentiment data) already partially compensated for class imbalance through strong lexical and pragmatic priors, raising the performance ceiling and leaving little residual imbalance for the GCN to correct. Second, graph sparsity in the training set severely limited the GCN's opportunity to propagate class-discriminative signals. Under these conditions, the null result reflects the data set and the backbone constraints rather than a failure of the core hypothesis that the conversation structure can inform sentiment classification under imbalance.

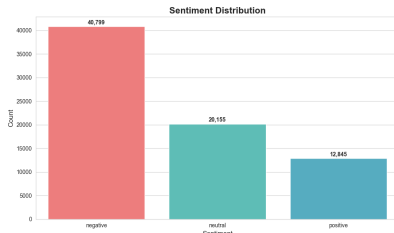


Fig. 2: Distribution of sentiment labels in the full dataset. The X-axis displays the three sentiment categories and Y-axis the frequency of sentiment class

B. Topic Modeling

Given that the proposed BertGCN extension did not improve over the plain RoBERTa baseline, the RoBERTa model was used to conduct inference over the remaining 69,335 unlabeled tweets. Similar to the findings of Aquino et al. [11], the negative sentiment class has the largest number of tweets as seen in Figure 2, with the neutral and positive classes trailing far behind. The domination of the negative sentiment class in this dataset reflects the findings of Schöne et al. [8], which highlights the further propagation of the negative sentiment in comparison to positive ones.

Topic coherence was evaluated using UMass and UCI metrics [26]. Both the UMass and UCI were calculated using the same 73,799 tweets dataset. For UMass, the study used a document-level co-occurrence with boolean document frequencies, while UCI used a sliding window of 10 tokens, justified by the short length of tweets (average ≈ 200 characters). Under this setup, the 5-cluster solution achieved the highest average UCI score (-1.913), outperforming 3 clusters (-2.473) and 4 clusters (-2.486). Although the 5-cluster solution yielded a lower (more negative) UMass score (-3.352) than the 3-cluster solution (-2.985), this study prioritized UCI for topic selection, as it is less sensitive to document frequency biases and more reflective of word association strength.

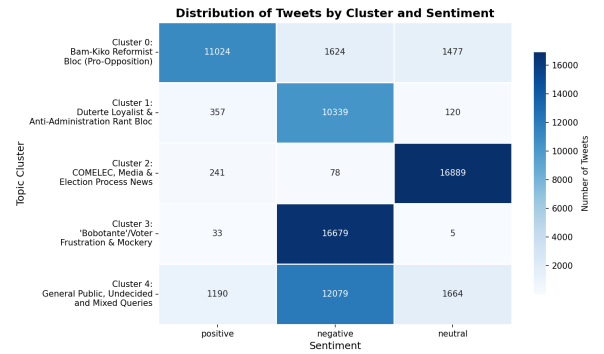


Fig. 3: Distribution of tweets per cluster and sentiment class. The X-axis displays the three sentiment categories and Y-axis the cluster categories

Figure 3 illustrates the sentiment class distribution per cluster, with cluster 0 being predominantly positive, cluster 2 negative, and clusters 1, 3, and 4 neutral. It is also determined that despite cluster 1 having the least amount of posts under it, most other clusters hover around the same number of posts. Figure 4 further clarifies the interpretation of these clusters by introducing the top 50 words present in each cluster. Based

on the LLR results, the following labels are derived from the top 100 words present in each cluster.

Cluster 0 is examined to be focused on the Pro-Opposition camp present in the current political sphere of the Philippines. This cluster is heavily dominated by keywords such as *bam*, *kiko*, *pangilinan*, *aquino*, *salamat*, *matalino*, *tama*, *tunay*, *good*, and *governance*. However, it does contain some negative labeled words such as *bobo*, *bakit*, and the like. Cluster 1, on the other hand, is titled as the Anti-Administration and Duterte Loyalist Bloc, as it features the keywords *Duterte*, *Sara*, *Digong*, *Bato*, *Robin*, *Villar*, *Bong*, *Imee*, *PDP*, *Quiboloy*, which are the names of key figures and organizations in that sector, which are mostly found in negative posts. Furthermore, it contains derogatory terms such as *bobo* and *tuta*, which further solidify its theme as a cluster laden with negative sentiment. The same negative dominant sentiment can be seen in Cluster 3, as it features more derogatory terms, which are identified as extremely characteristic of the cluster. Unlike cluster 1, however, this cluster is not endorsing candidates, but rather, is mocking or lamenting the quality of voters. Its theme can be represented as the general frustration of Filipinos with their fellow Filipino voters' intelligence, possibly from a more educated or cynical segment of social media.

On the other hand, cluster 2 tackles the more neutral side of the sentiment spectrum. Here, the top keywords follow along the lines of news and election processes announcements, which are seen in the following keywords: *elections*, *mayo*, *bilang*, *proklamasyon*, *automated*, *canvassers*, *disqualification*, *counting*, *online*, *source*, *detalye*, *ulat*, *radyopilipinas* (news outlet), *rp1news* (news outlet), *news5* (news outlet). There is no strong candidate endorsement except for low-weight candidate names. The tone is factual and news-oriented. Cluster 4 follows along the neutral dominant sentiment, but this time tackles the curiosity and queries made by the general public. This cluster has a mix of keywords pertaining to the election procedures, such as *commission*, *elections*, *2025*, *mayo*, *city*, *pnp* (Philippine National Police), *senado*. In addition, it also contains curiosity words like *sino*, *pwede*, *kahit*, *siguro*, and *iboboto*. It lacks the strong pro-BamKiko signal of Cluster 0 and the negative sentiment signal of Cluster 1. It appears to be undecided, or the general public is asking questions about candidates and processes.

It is also important to note that in clusters 0 and 1, obfuscated Twitter usernames were captured. Additionally, hashtags which were not used during the data collection, such as “#51pangilinankiko”, “#5aquinobam”, “#votewisely”, “#dapattotoo”, indicate that elements unique to the social media environment play an important role in online political discussion, as first posited by Aquino et al. [11].

V. CONCLUSION AND FUTURE WORK

This paper extended the EmoBertGCN framework of Aquino et al. [11] by incorporating conversation-based social edges into its graph structure and using tweets from the 2025 Philippine general midterm elections as the corpus. Despite the insignificant difference in performance between the proposed integration of the conversational structure present in social

media and the BertGCN framework, the integration provided meaningful insight into the limitations of the dataset used to implement it. Specifically, the number of conversation-connected posts was too sparse, constituting only 23.5% of the training set, which limited the conversational context's influence during the model's training. Furthermore, the domain-specialized RoBERTa pretrained specifically for code-switched Tagalog sentiment classification achieved a high metric that beat both the baseline BertGCN and the study's methodology, necessitating substantial improvements by both the baseline BertGCN and this study's methodology to demonstrate a significant contribution.

Notwithstanding the null result, the research still produced several contributions in both the field of NLP and in future analysis on online political discourse in the Philippines. A dataset of both manually and machine-annotated 73,799 tweets situated during a key electoral period has been produced, which serves as a resource for future NLP research on Tagalog and code-mixed text. Additionally, the adopted topic modeling from Aquino et al. [11]'s paper identified five interpretable discourse clusters: a Pro-Opposition cluster, an Anti-Administration and Duterte Loyalist cluster, an election news and COMELEC cluster, a voter frustration and mockery cluster, and a general public inquiry cluster.

For future research, it is recommended to expand the applications of these models on different social media platforms, while also taking into consideration the interaction structure unique to these platforms. Most importantly, it is advised to future researchers to collect as much conversation information as possible in order to confirm the impact of its integration on the graph structure of the BertGCN framework. The utilization of newer GNN architectures could be supplied to the framework rather than GCN, in order to account for the heterophily characteristic of political discourse [16], [27], [28]. Lastly, to ensure more reliable annotation of tweet sentiment, future researchers are recommended to explore more thorough guidance of experts. Together, these directions would enable a more complete evaluation of the theoretical premise that conversational context carries meaningful sentiment information for graph-based learning.

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Fig. 4: Word Clouds with Topic and LLR-based sentiment information of clusters. Words are colored based on the dominant sentiment of the documents they appear in most frequently: green for positive, red for negative, and grey for neutral.

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